

INTRODUCTION

MCQs benchmarks (e.g., GPQA, MMLU-Pro,...) are the most commonly used way to evaluate new LLMs capabilities.

Q: Capital of France?

- A. Berlin
- C. Paris
- B. Madrid
- D. Rome

But, they are running out of room.... !

PROBLEM STATEMENT

1

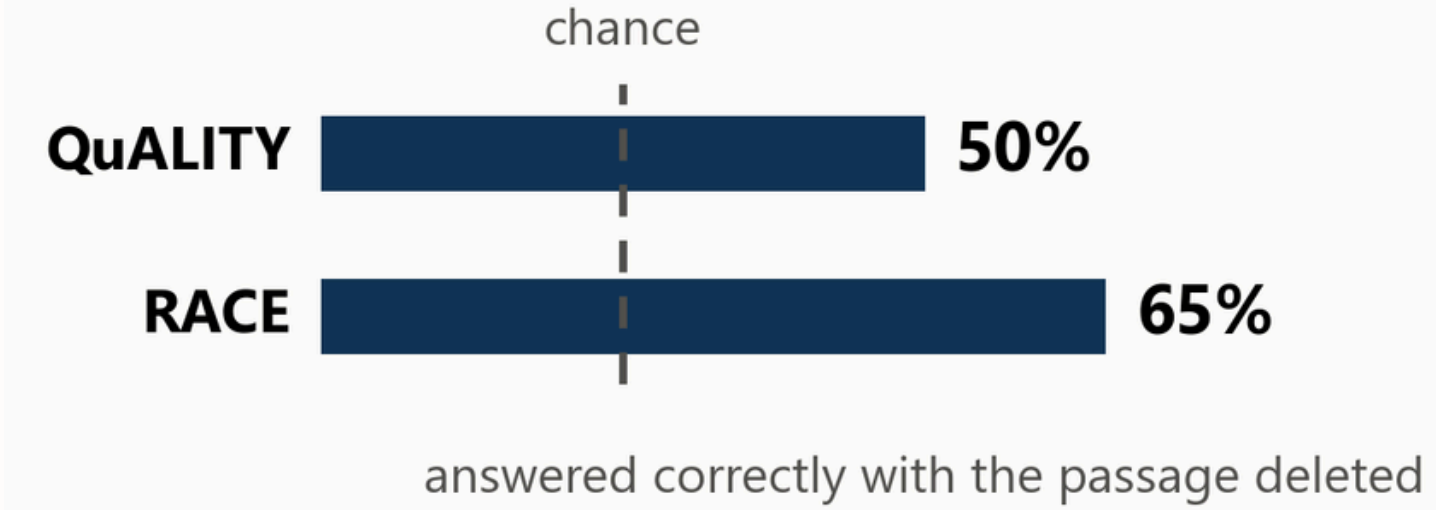
They saturate FAST!

Building a new benchmark takes months.
Models close the gap in weeks.



2

>50% of the questions never needed the input to answer correctly!

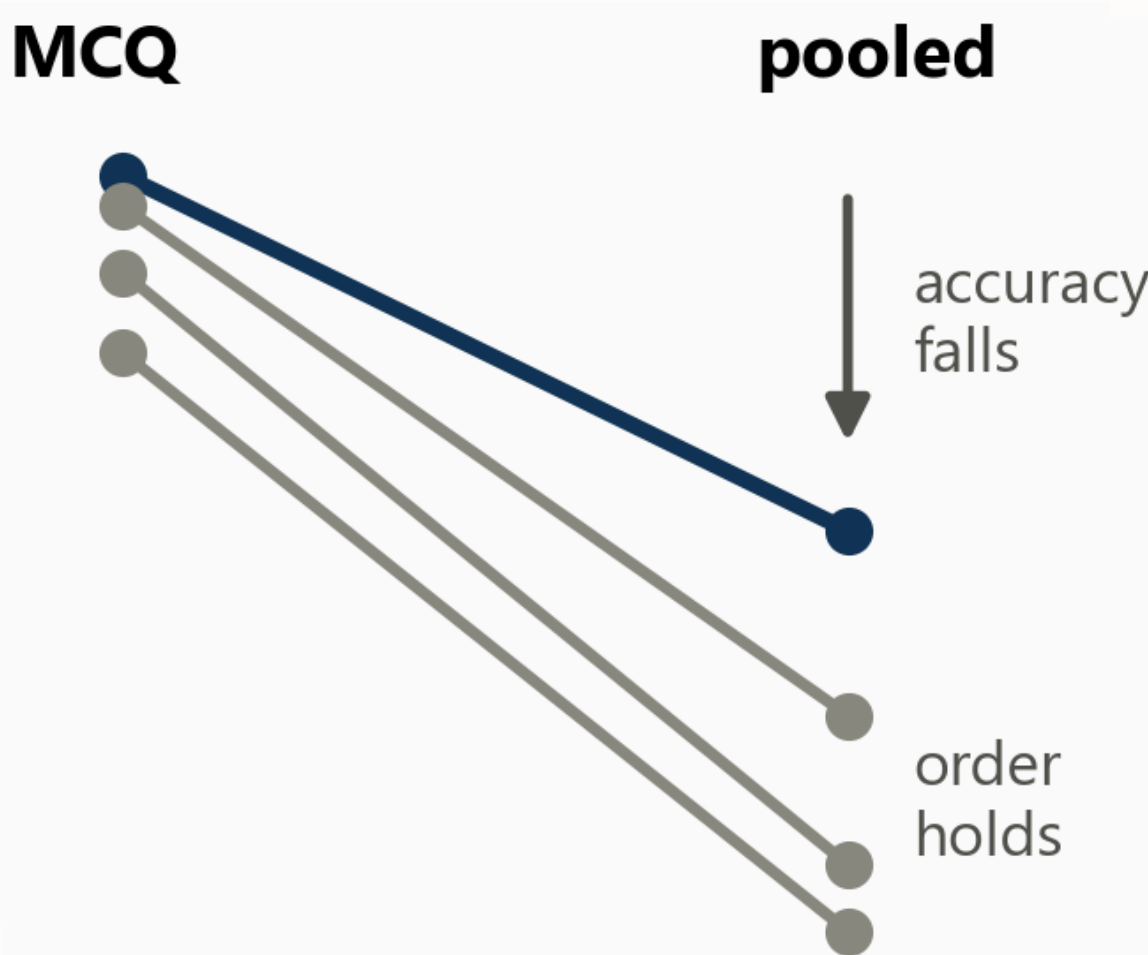


CONTRIBUTIONS

1

Benchmark de-saturation transform:

a de-saturation transform for any MCQ benchmark, with zero human work



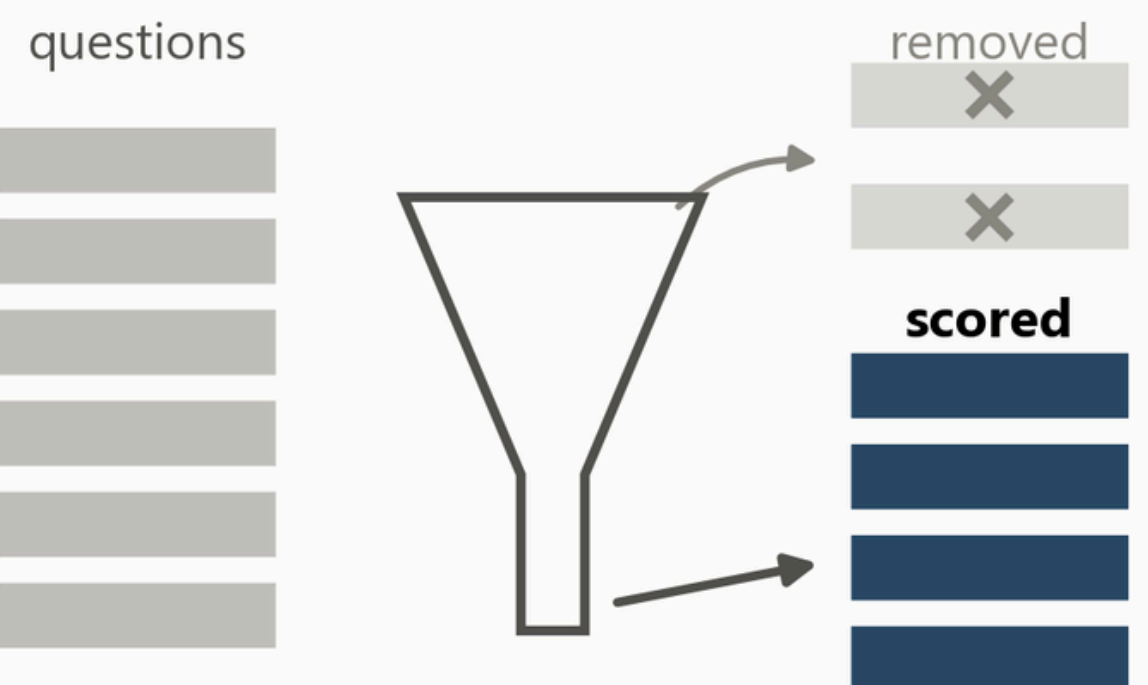
2

A filtering mechanism for MCQ benchmarks:

1) solvable without the context

2) ambiguous once pooled

3) separable by style



METHODOLOGY

Input (MCQs Benchmark)

Context (depends on the benchmark type)

Long context

Image

Video

...

Questions & their choices

Q1	A1	B1	C1	D1
Q2	A2	B2	C2	D2
Q3	A3	B3	C3	D3
Q4	A4	B4	C4	D4

1. Filtering
- Drop questions solvable without the context
- Q4 dropped
2. Withholding
- With a fixed ratio p, replace the correct choices with None
- Q2 → None
3. Pooling + Shuffling
- All choices of the kept questions

Our Formulation (Matching Task)

List A

Q1

Q2

Q3

List B

C2

B1

D3

A3

A2

C1

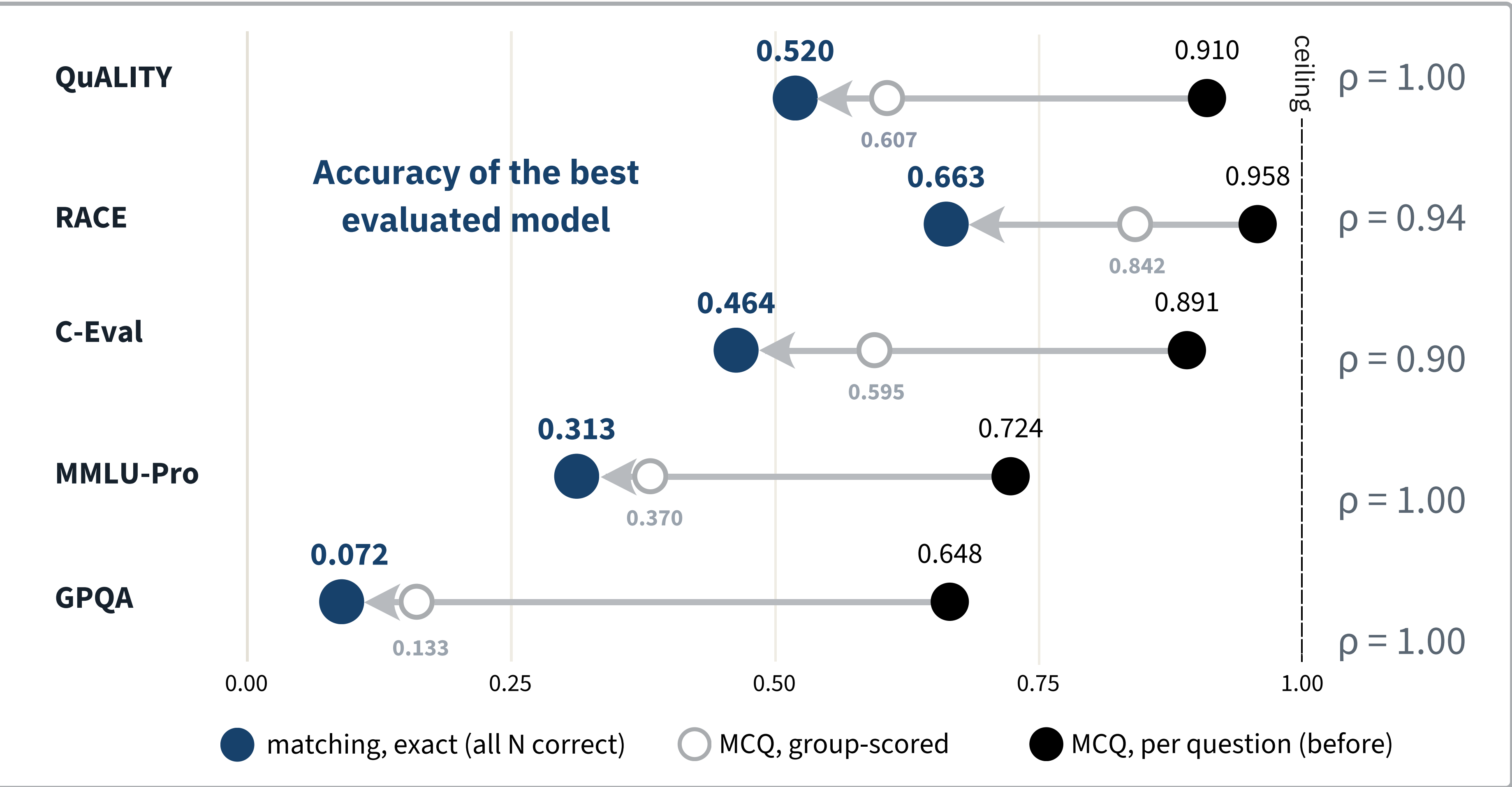
...

None

The model must assign **every** question in List A to exactly **one** entry in List B.

The transform. Filter out context-free questions, withhold a fraction p of golds, pool the rest into one shared list, and ask for the full assignment.

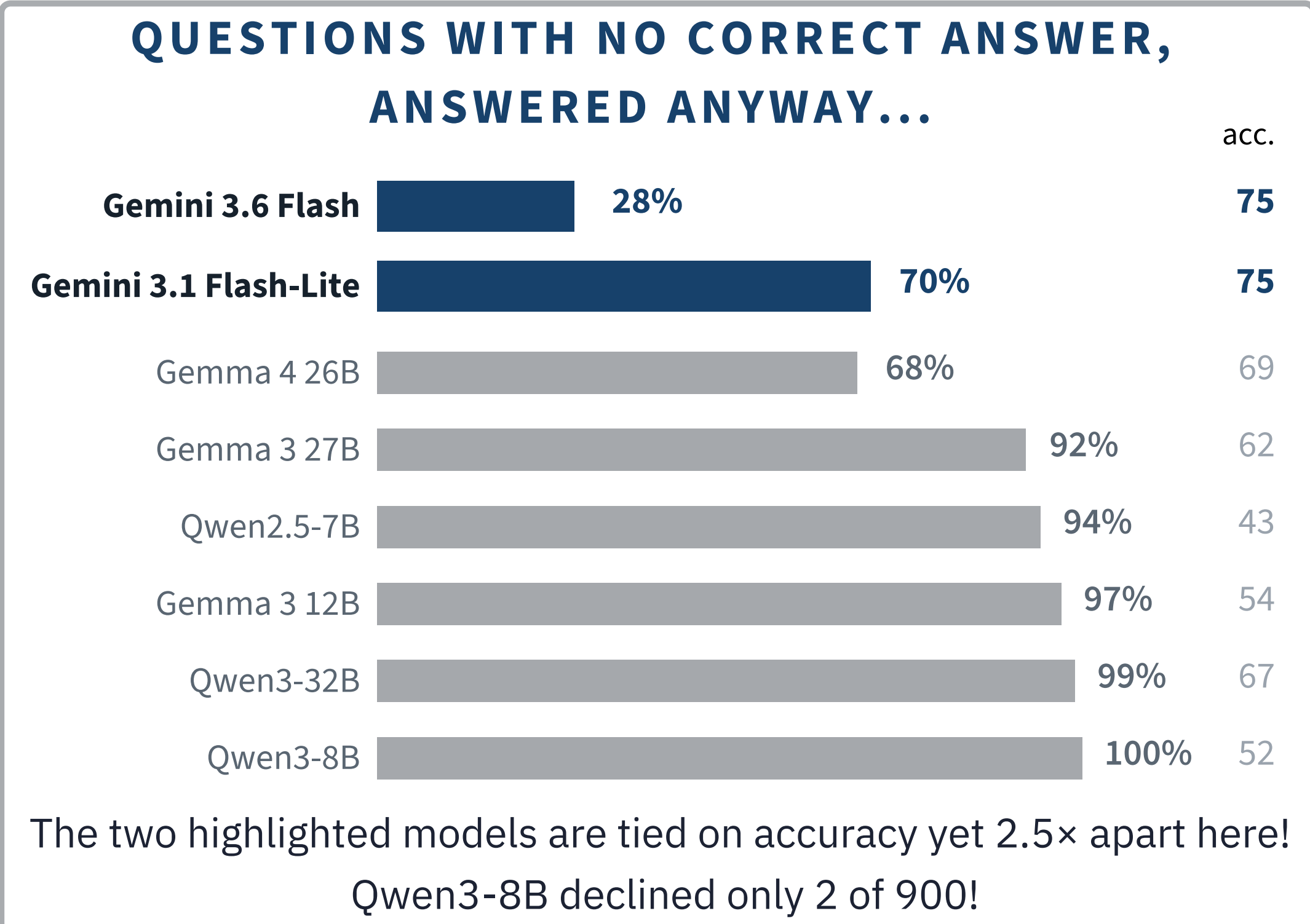
RESULTS (BENCHMARKS DESATURATION)



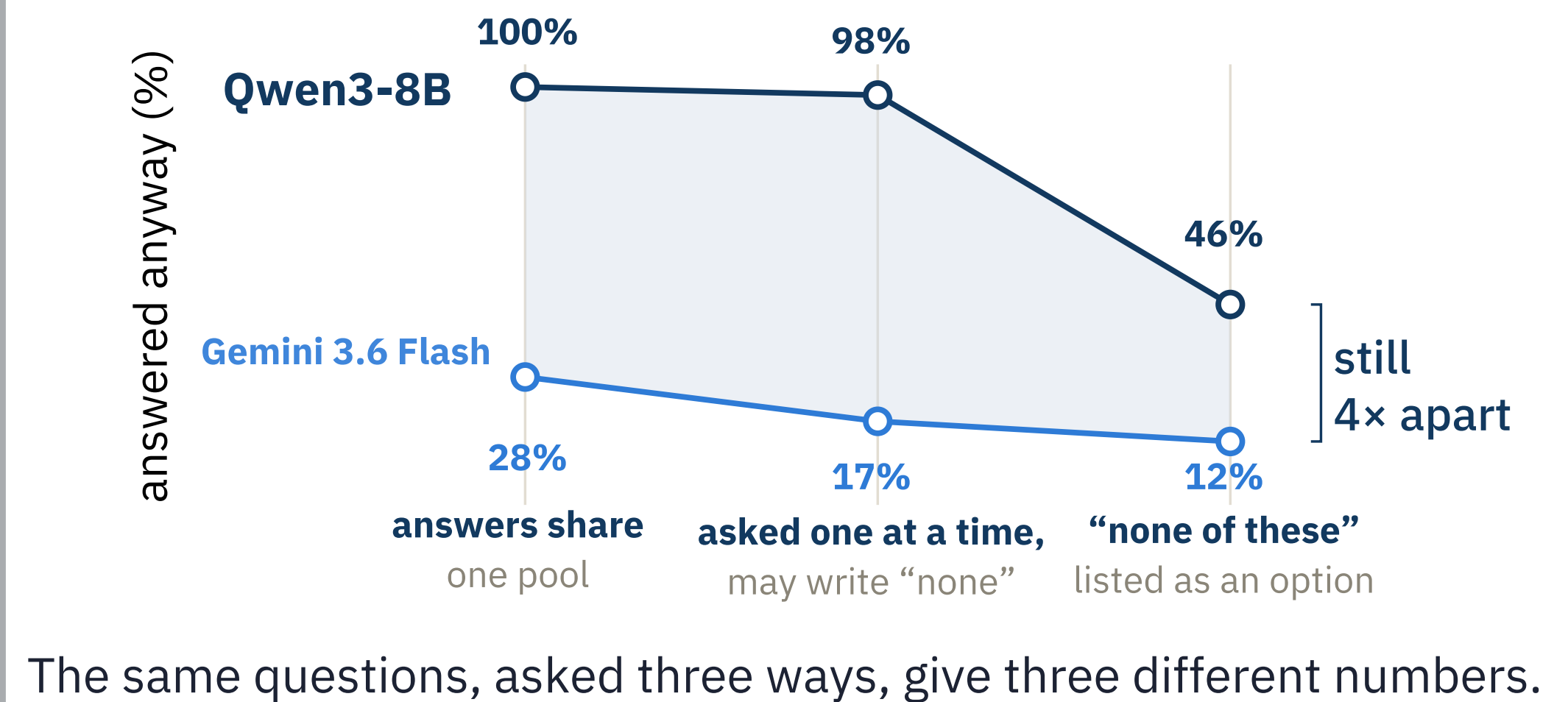
Accuracy of the best evaluated model before and after deasturation.

Group scoring removes part of the ceiling, matching removes the rest, and ρ (Spearman correlation with the original MCQ ranking) stays high across the benchmarks, proving the transform preserves the models rank.

RESULTS (ABSTENTION SCORES)



THE SAME MISSING QUESTIONS, ASKED THREE WAYS



WHAT TO DO WITH THIS?

- ➔ **Building a benchmark?**
Run the three screens before you release it.
- ➔ **Your benchmark saturating?**
Pool the answers. No new labels.
- ➔ **Reporting abstention?**
State the response format. It moves the number by 2×.

Resources:

